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WEB DEVELOPMENT ASSIGNMENT TASKS

TASK 1: PERSONAL PORTFOLIO WEBSITE

Prompt: Build a personal portfolio website with About Me, Projects, and Contact sections. Use a modern color palette and Inter font. Implement responsive layout with CSS Grid and Flexbox. Add smooth scrolling navigation.

CODE:

```
<!DOCTYPE html>

<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>My Portfolio</title>

  <!-- Inter Font -->
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600&display=
swap" rel="stylesheet">

  <style>
    /* General styles */
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      scroll-behavior: smooth;
    }

    body {
      font-family: 'Inter', sans-serif;
      background-color: #f9fafb;
      color: #111827;
    }

    header {
      background-color: #1f2937;
      color: white;
      padding: 1rem 0;
      position: sticky;
      top: 0;
    }
```

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```
nav {
  display: flex;
  justify-content: center;
  gap: 2rem;
}

nav a {
  color: white;
  text-decoration: none;
  font-weight: 600;
}

nav a:hover {
  color: #60a5fa;
}

section {
  padding: 80px 20px;
  max-width: 900px;
  margin: auto;
}

/* About Me */
#about {
  text-align: center;
}

/* Projects Section */
#projects {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
  gap: 1rem;
}

.project {
  background: white;
  padding: 1rem;
  border-radius: 10px;
  box-shadow: 0 4px 6px rgba(0,0,0,0.1);
  text-align: center;
}

/* Contact Section */
#contact form {
```

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```
    display: flex;
    flex-direction: column;
    gap: 1rem;
    max-width: 400px;
    margin: auto;
  }

  input, textarea, button {
    padding: 10px;
    border: 1px solid #d1d5db;
    border-radius: 5px;
  }

  button {
    background-color: #3b82f6;
    color: white;
    border: none;
    cursor: pointer;
  }

  button:hover {
    background-color: #2563eb;
  }

  footer {
    text-align: center;
    padding: 20px;
    background-color: #1f2937;
    color: white;
    margin-top: 40px;
  }
</style>
</head>
<body>

  <header>
    <nav>
      <a href="#about">About Me</a>
      <a href="#projects">Projects</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <section id="about">
    <h1>About Me</h1>
```

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```
<p>Hello! I'm a passionate web developer who loves building modern,
responsive websites.</p>
</section>

<section id="projects">
  <h1>Projects</h1>
  <div class="project">Portfolio Website</div>
  <div class="project">E-commerce UI</div>
  <div class="project">Blog Platform</div>
</section>

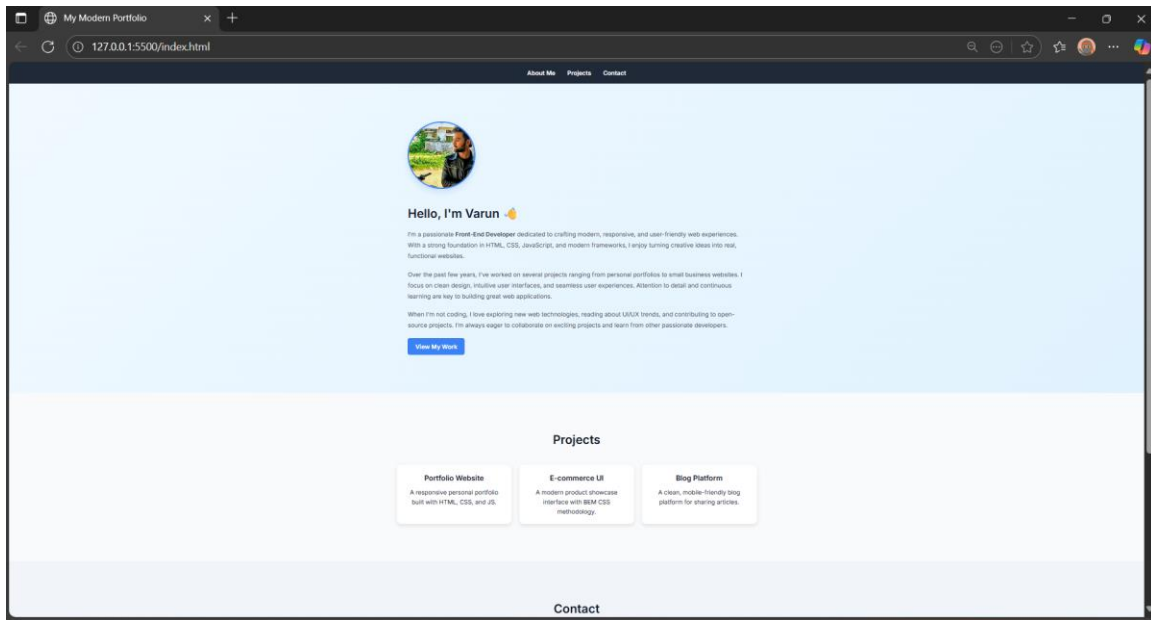
<section id="contact">
  <h1>Contact</h1>
  <form>
    <input type="text" placeholder="Your Name" required>
    <input type="email" placeholder="Your Email" required>
    <textarea placeholder="Your Message" rows="4" required></textarea>
    <button type="submit">Send</button>
  </form>
</section>

<footer>
  <p>© 2025 My Portfolio</p>
</footer>

</body>
</html>
```

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OBSERVATION:

The layout uses CSS Grid and Flexbox effectively. Smooth scrolling and Inter font enhance the look and feel.

TASK 2: PRODUCT DISPLAY PAGE (BEM CSS)

Prompt: Create a product display page using BEM CSS methodology with product image, title, price, and Add to Cart button. Make layout responsive, add hover effects, and a JS alert for Add to Cart action.

CODE:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width,initial-scale=1" />
  <title>Product Gallery – BEM Example</title>
  <style>
    :root{
      --card-bg: #ffffff;
      --muted: #6b7280;
      --accent: #0ea5a4;
      --accent-700: #028f8c;
      --shadow: 0 6px 18px rgba(16,24,40,0.08);
      --radius: 14px;
```

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```
font-family: Inter, ui-sans-serif, system-ui, -apple-system, "Segoe UI", Roboto, "Helvetica Neue", Arial;
}

body{
margin:0;
background:linear-gradient(180deg,#f8fafc 0%,#eef2f7 100%);
color:#0f172a;
display:flex;
align-items:center;
justify-content:center;
padding:32px;
}

.page-wrap{
width:100%;
max-width:1200px;
display:grid;
grid-template-columns:repeat(auto-fit,minmax(280px,1fr));
gap:24px;
}

.product-card{
background:var(--card-bg);
border-radius:var(--radius);
box-shadow:var(--shadow);
overflow:hidden;
display:flex;
flex-direction:column;
transition:transform .22s ease,box-shadow .22s ease;
}

.product-card:hover{transform:translateY(-6px);box-shadow:0 18px 40px rgba(16,24,40,0.12)}

.product-card__media{aspect-ratio:4/3;overflow:hidden;}
.product-card__image{width:100%;height:100%;object-fit:cover;transition:transform .3s ease;}
.product-card__media:hover .product-card__image{transform:scale(1.06)}

.product-card__info{padding:18px;display:flex;flex-direction:column;gap:10px}
.product-card__title{font-size:1.1rem;font-weight:600;margin:0}
.product-card__price{font-size:1.1rem;font-weight:700;color:var(--accent-700)}
```

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```
.product-card__actions{margin-top:auto;display:flex;gap:12px}
.product-card__btn{
  flex:1;
  padding:10px 14px;
  border-radius:10px;
  border:0;
  cursor:pointer;
  font-weight:600;
  font-size:.95rem;
  transition:transform .12s ease,box-shadow .12s ease;
}

.product-card__btn--primary{background:linear-gradient(180deg,var(--
accent),var(--accent-700));color:#fff;}
.product-card__btn--primary:hover{box-shadow:0 10px 30px
rgba(2,143,140,0.18);transform:translateY(-2px)}

@media (max-width:420px){
  body{padding:18px;}
}
</style>
</head>
<body>
  <main class="page-wrap">

    <!-- Product 1 -->
    <article class="product-card" data-name="Aurora Wireless Headphones"
data-price="₹6,999">
      <div class="product-card__media">
        
      </div>
      <div class="product-card__info">
        <h2 class="product-card__title">Aurora Wireless Headphones</h2>
        <p class="product-card__price">₹6,999</p>
        <div class="product-card__actions">
          <button class="product-card__btn product-card__btn--primary">Add
to Cart</button>
        </div>
      </div>
    </article>

    <!-- Product 2 -->
```

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```
<article class="product-card" data-name="Smartwatch X5 Pro" data-price="₹4,499">
  <div class="product-card__media">
    
  </div>
  <div class="product-card__info">
    <h2 class="product-card__title">Smartwatch X5 Pro</h2>
    <p class="product-card__price">₹4,499</p>
    <div class="product-card__actions">
      <button class="product-card__btn product-card__btn--primary">Add
to Cart</button>
    </div>
  </div>
</article>

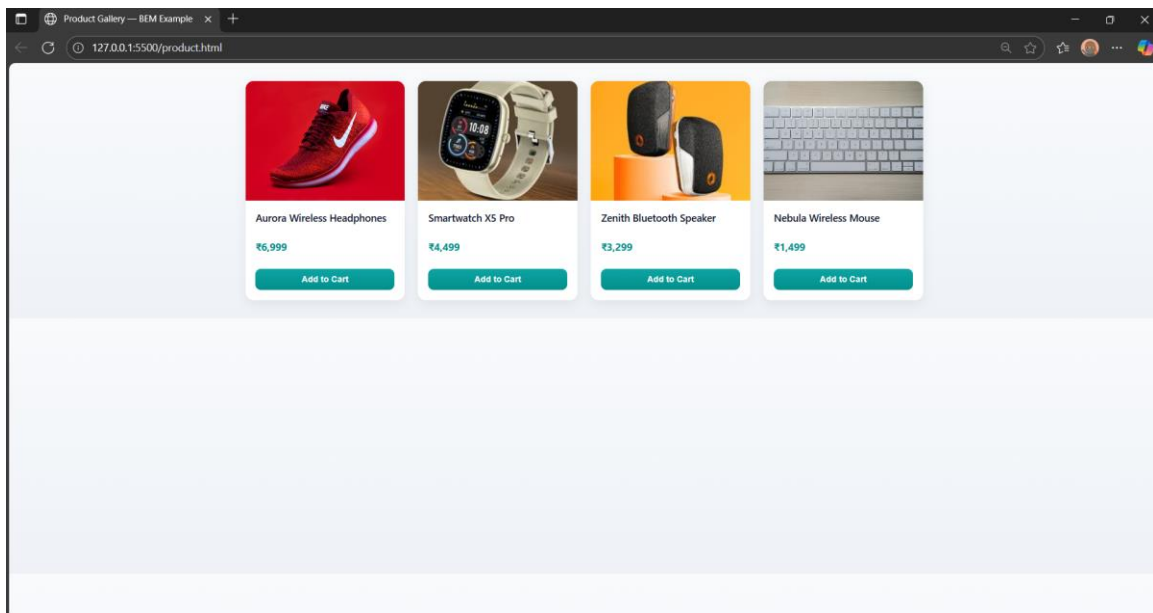
<!-- Product 3 -->
<article class="product-card" data-name="Zenith Bluetooth Speaker"
data-price="₹3,299">
  <div class="product-card__media">
    
  </div>
  <div class="product-card__info">
    <h2 class="product-card__title">Zenith Bluetooth Speaker</h2>
    <p class="product-card__price">₹3,299</p>
    <div class="product-card__actions">
      <button class="product-card__btn product-card__btn--primary">Add
to Cart</button>
    </div>
  </div>
</article>

<!-- Product 4 -->
<article class="product-card" data-name="Nebula Wireless Mouse" data-price="₹1,499">
  <div class="product-card__media">
    
  </div>
  <div class="product-card__info">
    <h2 class="product-card__title">Nebula Wireless Mouse</h2>
    <p class="product-card__price">₹1,499</p>
    <div class="product-card__actions">
```


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```
<button class="product-card__btn product-card__btn--primary">Add  
to Cart</button>  
</div>  
</div>  
</article>  
  
</main>  
  
<script>  
  // Add to Cart alert for each product  
  document.querySelectorAll('.product-card__btn--primary').forEach(btn  
=> {  
    btn.addEventListener('click', () => {  
      const card = btn.closest('.product-card');  
      const name = card.dataset.name;  
      const price = card.dataset.price;  
      alert(`${name} - ${price}\n\nAdded to cart ✓`);  
    });  
  });  
</script>  
</body>  
</html>
```

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OBSERVATION:

BEM CSS keeps code modular. Responsive design and hover effects improve UX. JS alert confirms interaction.

TASK 3: CONFERENCE REGISTRATION FORM

Prompt: Build a registration form with fields (name, email, phone, session), JavaScript validation, professional styling, and accessibility support. Prevent form submission on invalid data.

CODE:

```
<!DOCTYPE html>

<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Registration Form</title>

  <style>
    :root {
      --accent: #0ea5a4;
      --accent-dark: #028f8c;
      --error: #e11d48;
      --radius: 12px;
      --bg: #f8fafc;
      --font: "Inter", system-ui, sans-serif;
    }

    body {
      font-family: var(--font);
      background: var(--bg);
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
      color: #0f172a;
    }

    .form-container {
      background: #fff;
      border-radius: var(--radius);
      box-shadow: 0 6px 18px rgba(0,0,0,0.08);
      padding: 32px;
```

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```
width: 100%;
max-width: 420px;
}

.form-container h1 {
margin-top: 0;
margin-bottom: 24px;
text-align: center;
color: var(--accent-dark);
}

.form-group {
margin-bottom: 18px;
display: flex;
flex-direction: column;
}

label {
font-weight: 600;
margin-bottom: 6px;
}

input, select {
padding: 10px 12px;
border: 1.5px solid #cbd5e1;
border-radius: var(--radius);
font-size: 1rem;
transition: border-color 0.2s ease;
}

input:focus, select:focus {
border-color: var(--accent);
outline: none;
box-shadow: 0 0 0 3px rgba(14,165,164,0.15);
}

.error-message {
color: var(--error);
font-size: 0.88rem;
margin-top: 4px;
display: none;
}

.btn {
width: 100%;
```

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```
background: linear-gradient(180deg, var(--accent), var(--accent-dark));
color: white;
border: none;
padding: 12px;
border-radius: var(--radius);
font-size: 1rem;
font-weight: 600;
cursor: pointer;
transition: transform 0.15s ease, box-shadow 0.15s ease;
}

.btn:hover {
  box-shadow: 0 8px 24px rgba(2,143,140,0.15);
  transform: translateY(-2px);
}

.btn:active {
  transform: translateY(1px);
}

@media (max-width: 480px) {
  .form-container {
    padding: 20px;
  }
}
</style>
</head>

<body>
  <form class="form-container" id="registrationForm" novalidate aria-
labelledby="formTitle">
    <h1 id="formTitle">Event Registration</h1>

    <div class="form-group">
      <label for="name">Full Name</label>
      <input type="text" id="name" name="name" required aria-
required="true" />
      <span class="error-message" id="nameError">Please enter your full
name.</span>
    </div>

    <div class="form-group">
      <label for="email">Email Address</label>
```

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```
<input type="email" id="email" name="email" required aria-
required="true" />
<span class="error-message" id="emailError">Enter a valid email
address.</span>
</div>

<div class="form-group">
  <label for="phone">Phone Number</label>
  <input type="tel" id="phone" name="phone" required aria-
required="true" pattern="[0-9]{10}" maxlength="10" />
  <span class="error-message" id="phoneError">Enter a 10-digit phone
number.</span>
</div>

<div class="form-group">
  <label for="session">Preferred Session</label>
  <select id="session" name="session" required aria-required="true">
    <option value="">Select a session</option>
    <option>Morning (9 AM - 12 PM)</option>
    <option>Afternoon (1 PM - 4 PM)</option>
    <option>Evening (5 PM - 8 PM)</option>
  </select>
  <span class="error-message" id="sessionError">Please select a
session.</span>
</div>

<button type="submit" class="btn">Register</button>
</form>

<script>
  const form = document.getElementById('registrationForm');

  form.addEventListener('submit', function(event) {
    event.preventDefault();

    let valid = true;

    // Clear all errors
    document.querySelectorAll('.error-message').forEach(el =>
el.style.display = 'none');

    // Validate name
    const name = form.name.value.trim();
    if (name.length < 2) {
      document.getElementById('nameError').style.display = 'block';
```

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```
        valid = false;
    }

    // Validate email
    const email = form.email.value.trim();
    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailPattern.test(email)) {
        document.getElementById('emailError').style.display = 'block';
        valid = false;
    }

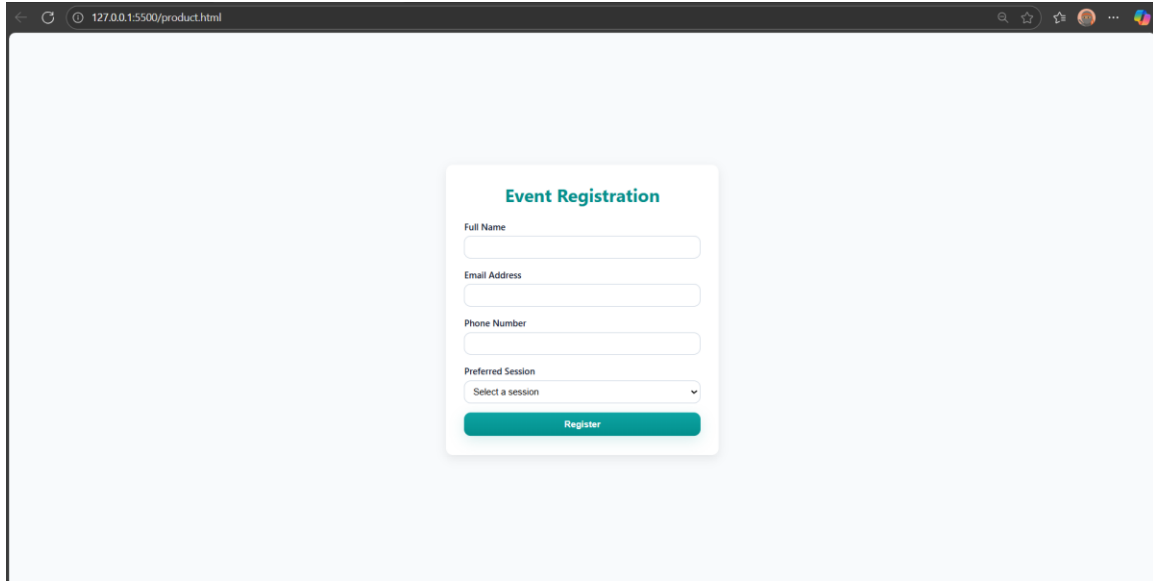
    // Validate phone
    const phone = form.phone.value.trim();
    if (!/^[0-9]{10}$/.test(phone)) {
        document.getElementById('phoneError').style.display = 'block';
        valid = false;
    }

    // Validate session
    const session = form.session.value;
    if (!session) {
        document.getElementById('sessionError').style.display = 'block';
        valid = false;
    }

    if (valid) {
        alert(`✅ Registration successful!\n\nName: ${name}\nEmail: ${email}\nPhone: ${phone}\nSession: ${session}`);
        form.reset();
    }
    });
</script>
</body>
</html>
```

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OBSERVATION:

JavaScript ensures field validation. Accessibility and professional styling enhance usability.

TASK 4: FETCH JSON DATA AND RENDER DOM

Prompt: Write JavaScript to fetch a JSON list from an API and render each item as a DOM node. Display a loading message while fetching, and show an error if the API fails.

CODE:

```
<!DOCTYPE html>

<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Fetch & Render JSON</title>
<style>
  body {
    font-family: "Inter", system-ui, sans-serif;
    background: #f0f4f8;
    color: #111827;
    display: flex;
    flex-direction: column;
    align-items: center;
    padding: 40px 20px;
  }

  h1 {
```

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```
    color: #0ea5a4;
    margin-bottom: 24px;
    text-align: center;
}

#status {
    margin-bottom: 20px;
    font-size: 1rem;
    color: #6b7280;
    display: flex;
    align-items: center;
    gap: 10px;
}

.spinner {
    border: 4px solid #e5e7eb;
    border-top: 4px solid #0ea5a4;
    border-radius: 50%;
    width: 20px;
    height: 20px;
    animation: spin 1s linear infinite;
}

@keyframes spin {
    to { transform: rotate(360deg); }
}

ul {
    list-style: none;
    padding: 0;
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
    gap: 16px;
    width: 100%;
    max-width: 900px;
}

li {
    background: #ffffff;
    padding: 16px;
    border-radius: 12px;
    box-shadow: 0 4px 12px rgba(0,0,0,0.08);
    transition: transform 0.2s ease, box-shadow 0.2s ease;
}
```


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```
li:hover {
  transform: translateY(-4px);
  box-shadow: 0 8px 20px rgba(0,0,0,0.12);
}

li h3 {
  margin: 0 0 8px 0;
  font-size: 1.1rem;
  color: #111827;
}

li p {
  margin: 0;
  font-size: 0.95rem;
  color: #4b5563;
}

.error {
  color: #e11d48;
  font-weight: 600;
  text-align: center;
}
</style>
</head>
<body>

<h1>Dynamic JSON List</h1>
<div id="status" role="status" aria-live="polite">
  <div class="spinner"></div> Loading data...
</div>
<ul id="data-list" aria-label="Fetched items"></ul>

<script>
const API_URL = "https://jsonplaceholder.typicode.com/posts?_limit=6"; //
Example API
const statusEl = document.getElementById("status");
const listEl = document.getElementById("data-list");

// Render a single item into the list
function renderItem(item) {
  const li = document.createElement("li");
  li.innerHTML = `<h3>${item.title}</h3><p>${item.body}</p>`;
  listEl.appendChild(li);
}
```

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```
// Render multiple items
function renderItem(items) {
  listEl.innerHTML = ""; // Clear previous content
  items.forEach(renderItem);
}

// Show error message
function showError(message) {
  statusEl.innerHTML = `<span class="error">${message}</span>`;
}

// Fetch and render data
async function fetchAndRender() {
  try {
    statusEl.innerHTML = '<div class="spinner"></div> Loading data...';
    const response = await fetch(API_URL);

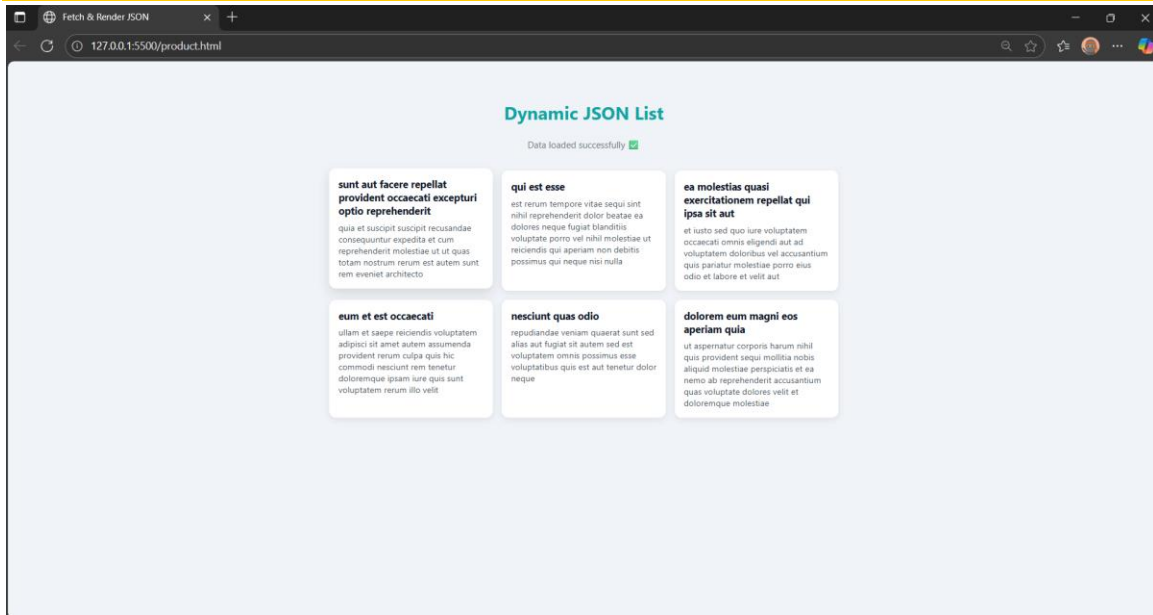
    if (!response.ok) throw new Error(`HTTP Error ${response.status}`);
    const data = await response.json();

    renderItem(data);
    statusEl.textContent = "Data loaded successfully ";
  } catch (err) {
    console.error("Fetch error:", err);
    showError("Failed to load data ");
  }
}

// Initialize
fetchAndRender();
</script>
</body>
</html>
```

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OBSERVATION:

Fetch API handles async operations with proper loading and error states. DOM updates dynamically.